HARSHITA SHARMA

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OBJECTIVE

AI/ML Engineer with hands-on experience building and deploying Computer Vision, Machine Learning, and Generative AI systems (LLMs, RAG, chatbots). Skilled in Python and the end-to-end ML lifecycle — from data preprocessing and model training to real-time inference pipelines and API integration. Experienced in designing scalable, low-latency AI solutions and translating data into actionable insights. Eager to build production-grade AI applications.

EDUCATION

Bachelor of technology in Computer Science Engineering with specialization in AIML, Abdul Kalam Technical University (2022- 2026)
Relevant Coursework: Artificial Intelligence, Machine Learning and Data Science.

SKILLS

Programming Languages	Python , SQL ,C++
ML	Supervised and Unsupervised Learning, Model Training, Evaluation, Optimization , Feature Engineering.
Gen AI and LLM	LLMs, RAG, Prompt Engineering, AI Agents, LangChain, Vector DBs (FAISS, ChromaDB), Chatbot Development, NLP.
LLM API and Models	OpenAI (GPT), Google Gemini, Anthropic Claude, Groq, LLama, Hugging Face Transformers.
Computer Vision	YOLOv8, OpenCV, Object Detection, Video Analytics, Real time inference.
Data Science	Data Preprocessing, Data Visualization, Feature Engineering, Probability and Statistics.
Frameworks and Libraries	TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib
Tools and Deployment	REST APIs, FastAPI, Streamlit, Power BI, Tableau, Git, GitHub, Linux, AI Developer Tools (Claude Code), Google Colab.
Soft Skills	Leadership, Problem solving, Communication, Team work and Collaboration.

EXPERIENCE

AI Intern Jan 2026 - Present
AI Powered Video Surveillance and Anomaly Detection System *Codeacious Technologies, Noida*

- Developed computer vision and deep learning models using YOLOv8 and OpenCV for real-time object detection in video surveillance.
- Designed a multi-stage AI inference pipeline with motion-based frame filtering, object detection, and Visual LLM validation, reducing false positives and improving detection accuracy via event debouncing.
- Built scalable video processing pipelines for RTSP streams using multi-threading, producer-consumer architecture, and distributed inference, enabling reliable real-time monitoring with auto-reconnect mechanisms.

AIML Intern July 2025 - Oct 2025
LLM-Powered Mental Health Conversational AI Chatbot *Qmansys Infosolutions ,Uttar Pradesh*

- Developed an AI powered mental health chatbot leveraging LLMs like Langchain and Groq to provide empathetic, context aware responses and emotional support.
- Utilized Natural Language Processing(NLP) for intent detection, sentiment analysis and adaptive conversation flow.
- Optimized for scalability and low latency on Google Colab, improving accessibility for diverse user groups including visually impaired individuals. (Try it here)

DELOITTE DATA ANALYTICS JOB SIMULATION Oct 2025

- Completed a Deloitte job simulation involving data analysis and forensic technology..
- Created a data dashboard using Tableau
- Used Excel to classify data and draw business conclusions. (Link)

PROJECTS

OmniGuard AI — Real-time Hybrid RAG Anomaly Detection System for LLM Failures:

1. Architected a hybrid RAG pipeline pairing fast vector-similarity search with an LLM-as-judge path, lifting detection (F1 0.87 → 0.93) while routing only ~18% of requests to the LLM (≈80% cost cut).
2. Built a custom vector database with kNN semantic search over sentence-transformer embeddings and a RAG retrieval layer grounding LLM verdicts in the top-5 similar examples — served via async FastAPI with WebSocket real-time monitoring.
3. Productionized with Docker Compose, PostgreSQL audit logging, Redis caching (15× speedup), and GitHub Actions CI/CD at 70% test coverage.

Next Word Predictor:

1. Developed an LSTM-based NLP model for sequence prediction and text generation.
2. Performed text preprocessing, tokenization, and embedding, enabling contextual language understanding.
3. Demonstrated concepts of deep learning, sequence modeling, and generative AI systems (Try it here)

Medical Insurance Cost Prediction:

1. Conducted Exploratory Data Analysis (EDA) and data preprocessing on structured datasets.
2. Built regression models and evaluated using RMSE and R^2 metrics.
3. Extracted actionable insights from data to support data-driven decision making (Try it here)

CERTIFICATIONS

- TATA Data Visualisation Job Simulation (Link)
- GEN AI by AWS – Jan'24 (Link)
- AZURE AI SERVICES by MICROSOFT - Dec'24 (Link)
- NLP by MICROSOFT - Jan'25 (Link)

LEADERSHIP AND ACHIEVEMENTS

- Developed and deployed web applications using python.

- Earned Industry recognized certifications for AI and Data Science.
- Mentored juniors in Python, Machine Learning, and Data Visualization during internal workshops, improving their project performance and coding confidence.

EXTRA CURRICULAR ACTIVITIES

- Represented college volleyball team in Inter-University Tournament and secured RUNNER UP position in 2025 championship.